

Lecture 01: The Learning Problem

Comprehensive Technical & Conceptual Guide

Based on Caltech CS156: “Learning from Data” by Prof. Yaser Abu-Mostafa

1 Introduction & Course Philosophy

Machine learning is a broad discipline spanning from highly abstract mathematical theory to extreme practice (rules of thumb and heuristics). The inclusion of topics in Caltech’s CS156 course is strictly guided by relevance to real-world learning systems.

- **Mathematical Foundations:** Provide the conceptual framework necessary to understand whether and how a machine can learn from data.
- **Practical Aspects:** Offer algorithmic tools and engineering techniques to build, debug, and validate actual learning systems.
- **Course Progression:** The storyline begins with foundational concepts and theoretical bounds (*theory/red phase*), establishing whether learning is feasible, before progressing to practical algorithms and validation techniques (*practice/blue phase*).

2 The Essence of Machine Learning

To determine if machine learning is applicable to a specific problem in any domain, three essential criteria must be present:

The Three Pillars of Machine Learning Applicability

1. **A Pattern Exists:** There must be an underlying relationship or structure connecting the inputs to the target output.
2. **Cannot Pin It Down Mathematically:** The underlying relationship cannot be easily expressed as a closed-form analytical equation (e.g., a 17th-order polynomial).
3. **Data is Available:** We possess historical data capturing instances of the pattern from which a model can learn.

2.1 Real-World Benchmark: The Netflix Movie Recommendation Problem

A prime illustration of machine learning’s commercial and technical value is the Netflix Prize competition, which awarded \$1,000,000 to the first team achieving a 10% improvement over Netflix’s in-house recommendation engine.

- **Commercial Impact:** Spot-on movie recommendations significantly increase user engagement and rental volume, generating revenue far exceeding the \$1M prize.
- **Latent Factor Model Intuition:** Viewers and movies can be represented as vectors in a d -dimensional factor space (e.g., $d \approx 300$ attributes capturing comedy preference, action level, blockbuster interest, lead actor preference, etc.).
- **Reverse Engineering via Learning:** Starting from random factor vectors for every user and movie, the learning algorithm iteratively updates parameters across 100 million historical ratings to align inner products $\mathbf{x}_{\text{user}}^T \mathbf{y}_{\text{movie}}$ with actual ratings.

3 Formalization of the Machine Learning Problem

To formalize the learning problem mathematically, we use the canonical metaphor of **Credit Approval** at a bank.

3.1 Mathematical Definitions

- **Input Vector** ($\mathbf{x} \in \mathcal{X}$): A d -dimensional vector representing instance attributes (e.g., age, salary, outstanding debt, years in residence). $\mathcal{X} = \mathbb{R}^d$ is the input space.
- **Target Output** ($y \in \mathcal{Y}$): The ground-truth outcome or decision. For binary credit approval, $\mathcal{Y} = \{-1, +1\}$ (where $+1$ represents credit extended/good customer, and -1 represents credit denied/bad customer).
- **Target Function** ($f : \mathcal{X} \rightarrow \mathcal{Y}$): The ideal, unobservable credit approval formula that dictates the true outcome. In practice, f is completely unknown to us.
- **Training Data** ($\mathcal{D}$): A historical dataset of N observations:

$$\mathcal{D} = \{(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_N, y_N)\} \quad (1)$$

- **Hypothesis Set** ($\mathcal{H}$): A set of candidate functions $h \in \mathcal{H}$ (where $h : \mathcal{X} \rightarrow \mathcal{Y}$) from which the algorithm selects a solution.
- **Learning Algorithm** ($\mathcal{A}$): An optimization procedure that searches $\mathcal{H}$ using dataset $\mathcal{D}$ to select the best hypothesis $g \in \mathcal{H}$.
- **Final Hypothesis** ($g \approx f$): The operational formula produced by the algorithm to approximate the unknown target function f on unseen future data.

3.2 The Role of the Hypothesis Set $\mathcal{H}$

Why explicitly restrict the learning algorithm to search within a predefined hypothesis set $\mathcal{H}$?

1. **No Practical Downside:** Every practical model implementation (Linear Models, Neural Networks, Support Vector Machines) inherently restricts candidate functions to a specific parametric family.
2. **Essential Theoretical Upside:** The mathematical properties (complexity, degrees of freedom) of $\mathcal{H}$ dictate whether learning is feasible and govern the generalization error on unseen data.

4 The First Concrete Model: The Perceptron

The Perceptron is a simple linear binary classification model that computes a weighted sum of input attributes and compares it to a decision threshold.

4.1 Formulation and Artificial Coordinate Trick

Given attributes $x_1, x_2, \dots, x_d$ and weights $w_1, w_2, \dots, w_d$:

$$\text{Credit Score} = \sum_{i=1}^d w_i x_i \quad (2)$$

Comparing the credit score to a threshold T :

$$h(\mathbf{x}) = \text{sign} \left(\sum_{i=1}^d w_i x_i - T \right) \quad (3)$$

By defining an artificial input coordinate $x_0 = +1$ and an artificial weight $w_0 = -T$, we unify the threshold into the weight vector:

$$h(\mathbf{x}) = \text{sign} \left(\sum_{i=0}^d w_i x_i \right) = \text{sign}(\mathbf{w}^T \mathbf{x}) \quad (4)$$

where $\mathbf{w} = [w_0, w_1, \dots, w_d]^T$ and $\mathbf{x} = [x_0, x_1, \dots, x_d]^T$.

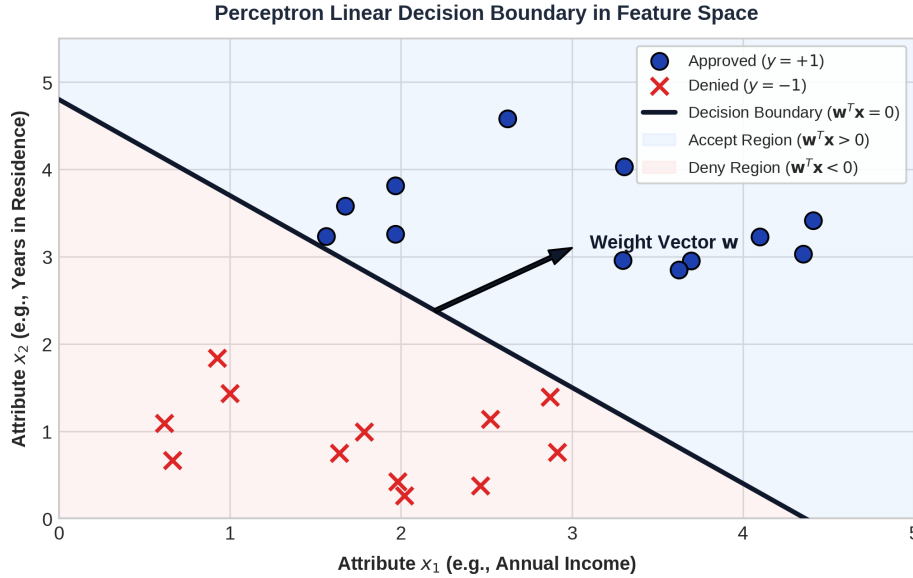


Figure 1: Perceptron Decision Boundary $\mathbf{w}^T \mathbf{x} = 0$ separating feature space into $+1$ (Approve) and -1 (Deny) regions.

4.2 The Perceptron Learning Algorithm (PLA)

For a dataset $\mathcal{D}$, PLA initializes weights (e.g., to $\mathbf{w} = \mathbf{0}$ or random values) and iteratively corrects misclassified points.

Perceptron Learning Algorithm (PLA) Step-by-Step

1. **Identify Misclassification:** Find a data point $(\mathbf{x}_n, y_n) \in \mathcal{D}$ such that:

$$\text{sign}(\mathbf{w}(t)^T \mathbf{x}_n) \neq y_n \quad (5)$$

2. **Update Weight Vector:** Adjust the weights using the rule:

$$\mathbf{w}(t+1) = \mathbf{w}(t) + y_n \mathbf{x}_n \quad (6)$$

3. **Repeat:** Cycle through data points until no misclassifications remain.

4.2.1 Geometric & Algebraic Intuition of Update Rule

- If $y_n = +1$, the model predicted negative ($\mathbf{w}^T \mathbf{x}_n < 0$). Updating $\mathbf{w} \leftarrow \mathbf{w} + \mathbf{x}_n$ increases the dot product $(\mathbf{w} + \mathbf{x}_n)^T \mathbf{x}_n = \mathbf{w}^T \mathbf{x}_n + \|\mathbf{x}_n\|^2$, pulling the score toward $+1$.
- If $y_n = -1$, the model predicted positive ($\mathbf{w}^T \mathbf{x}_n > 0$). Updating $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{x}_n$ decreases the dot product, pushing the score toward -1 .

4.2.2 Convergence Theorem & Limitations

- **Convergence Theorem:** If dataset $\mathcal{D}$ is **linearly separable**, PLA is guaranteed to converge in a finite number of steps to a hyperplane that perfectly separates all points.
- **Non-Separable Case:** If data contains noise or non-linear patterns, standard PLA oscillates infinitely without terminating. The **Pocket Algorithm** modifies PLA by keeping the best weight vector in a “pocket” throughout iterations.

5 Taxonomy of Machine Learning

Machine learning paradigms are broadly categorized based on the nature and availability of supervision in the training data:

Table 1: Taxonomy of Machine Learning Paradigms

Paradigm	Training Data Structure	Illustrative Example
Supervised Learning	Explicit pairs $(\mathbf{x}_n, y_n)$ containing both input and target output.	Coin classification with known denomination labels; credit approval with past payment history.
Unsupervised Learning	Unlabeled inputs $\mathbf{x}_n$ only; no target output y_n provided.	Coin clustering by size/mass into natural groups; customer segmentation; feature representation.
Reinforcement Learning	Input $\mathbf{x}_n$, trial output a_n , and a retroactive score/reward r_n .	Playing games (e.g., Backgammon, Chess); toddler learning not to touch hot teacups via pain response.

6 A Fundamental Learning Puzzle & Theoretical Dilemma

To demonstrate the core challenge of learning, consider a 3×3 binary pixel pattern grid (9-bit input vectors).

6.1 The Puzzle Experiment

- A training set $\mathcal{D}$ presents 6 pixel patterns mapped to $y \in \{-1, +1\}$.
- An unlabeled test pattern is presented to learners.
- **Outcome:** Different logical hypotheses (e.g., top-left pixel is white vs. pattern is symmetric) fit all 6 training points perfectly, yet yield completely opposite predictions (+1 vs. -1) on the test pattern.

6.2 Philosophical Takeaway: Learning vs. Memorization

The Fundamental Dilemma of Generalization

Outside the observed training set $\mathcal{D}$, the target function f remains completely unrestricted and unknown. Perfect performance on $\mathcal{D}$ alone represents **memorization**. Achieving **generalization** (predicting f outside $\mathcal{D}$) requires formal theoretical guarantees restricting the complexity of $\mathcal{H}$.

7 Key Lecture Q&A Insights

1. **Sampling Bias:** In credit approval, historical data only records individuals who were *accepted* for credit. The bank cannot directly observe outcomes for rejected applicants, introducing potential sampling bias that learning theory must address.

2. **Data Quantity vs. Model Complexity:** Generalization depends on having sufficient data points N relative to the degrees of freedom (complexity) of the hypothesis set $\mathcal{H}$. Fitting high-dimensional models (e.g., $7,000^{\text{th}}$ -order polynomials) on small datasets leads to severe overfitting.
3. **Data Snooping / Contamination:** Looking at dataset $\mathcal{D}$ before choosing $\mathcal{H}$ contaminates validation bounds and leads to overfitting noise rather than learning true underlying patterns.